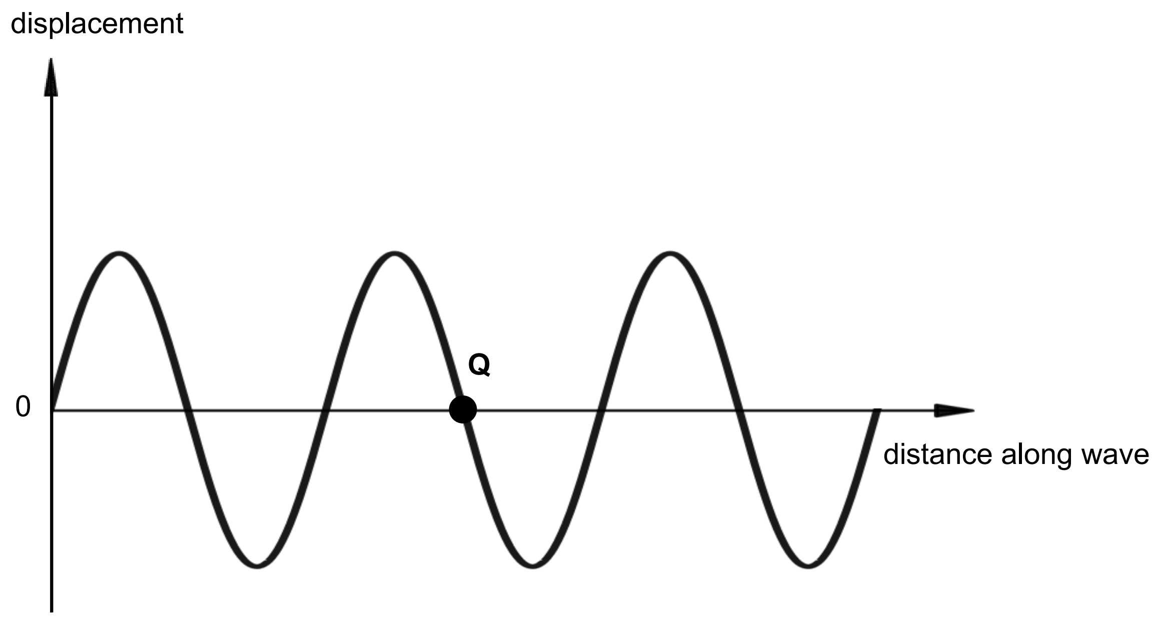
**15** This diagram shows the displacement-distance graph of a transverse wave at time, $t = 0$. Taking upwards as positive, point Q is a point on the wave and is travelling downwards at $t = 0$. Another point P is $\frac{7}{8}$ of a wavelength from point Q.

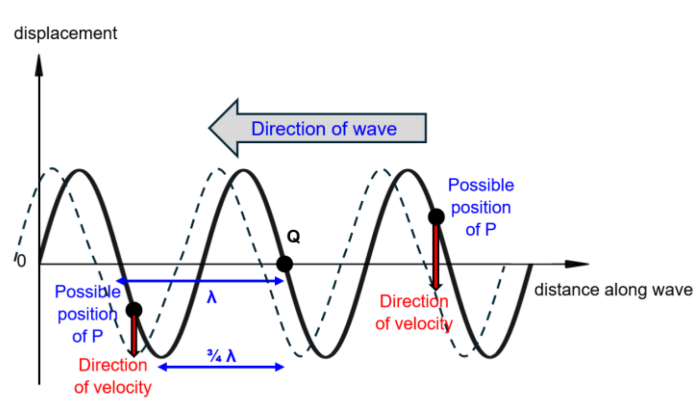



Which of the following descriptions shows a possible displacement and direction of velocity of point P?

| | displacement | direction of velocity |
|---|---|---|
| **A** | positive | positive |
| **B** | positive | 0 |
| **C** | 0 | negative |
| **D** | negative | negative |

Ans: D

Since the point Q is travelling downwards at $t = 0$, it implies that the transverse wave is travelling towards the **left**.




Since P is $\frac{7}{8}$ of a wavelength away from Q, there are 2 possible positions it can be at as shown above. Some time later, the wave profile will be as shown by the dotted line. In order for this to happen, if P is on the left side of Q, it must move downwards and have a negative displacement from equilibrium at $t = 0$. On the other hand, if P is on the right side of Q, it must move downwards and have a positive displacement from equilibrium at $t = 0$.

**16** The graph below shows the variation with time $t$ of the displacement $x$ of a simple harmonic